

ME F211 - MF F211 - Mechanics of Solids

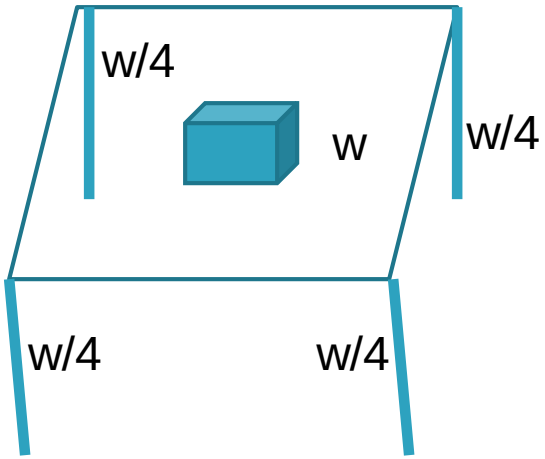
Fundamental principles of mechanics

Dr. Sharad Shrivastava
Department of Mechanical Engineering

BITS Pilani
Pilani Campus

Mechanics

Study of forces & motion



Mechanics of solids

Applied mechanics (to system of particular interest)

What comes under mech. of solids ?

We apply law of mechanics on structural elements for analysing & designing various machines and load bearing structure

- We used principle of mechanics to find support reactions
- In transfering these forces from their point of application to supports, materials of structure (wood/steel) develops: resisting forces, generates stresses, undergoes deformation (all these are studied in mech sol.)

Generalized Procedure in applied mechanics

Step1: Select system of interest.

Step2: Idealization of the system(simplification)

Step3: Apply principle of mechanics (predictions)

Step4: Compare these predictions with the actual system (tests and experiments)

Step5: If results not satisfactory construct a different idealized model and repeat the steps

In Research investigations step 1 and 2 are the most difficult ones

Step 3- needs understanding of forces and motions

Two different types of motions are possible in mechsol.

Overall change in position with time

↓

dynamics

Local distortions in shape: deformation

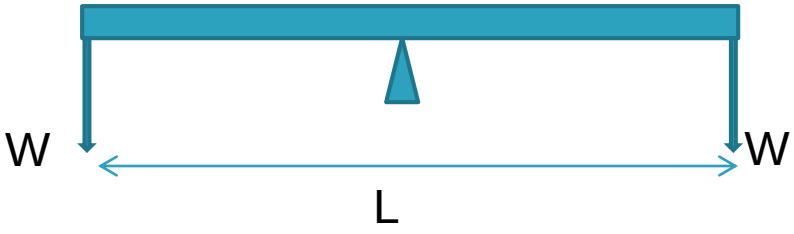
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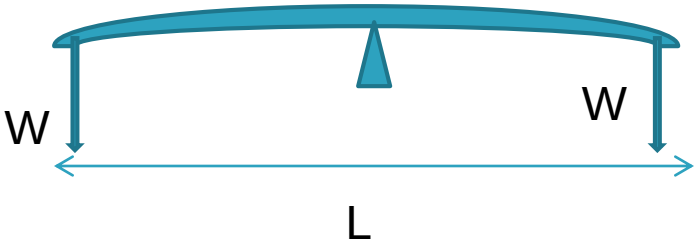
We will be considering static and rigid body (small deformation)

Rigid Bodies

No deformation (distance between two points of the body does not change under the action of applied force)

No Engg. structure practically is absolute rigid





➢As bending deformation is very small, the distance b/w each weight and fulcrum remains L/2. Hence it can be assumed rigid and straight. (if equlibrium of lever is to be studied then these assumptions are appropriate.

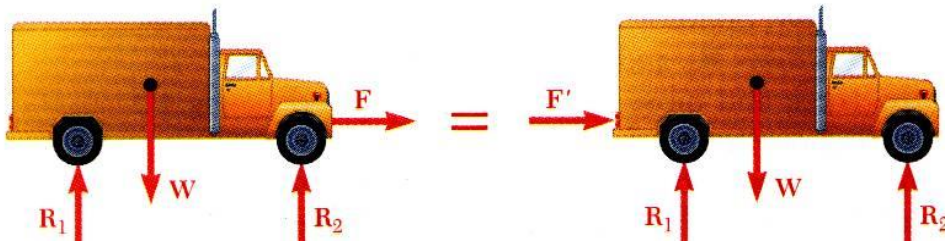
➢*But if strength of lever is to be studied then the small deformation has to be considered*

Principle of mechanics (law of transmissibility)

The point of application of a force may be transmitted along its line of action, without changing the effect of the force on any rigid body to which it may be applied.

The use of theorem is limited to those problems of statics in which we are interested only in condition of equilibrium of a rigid body and not with the internal forces to which it is applied

- Moving the point of application of the force **F** to the rear bumper does not affect the motion or the other forces acting on the truck.

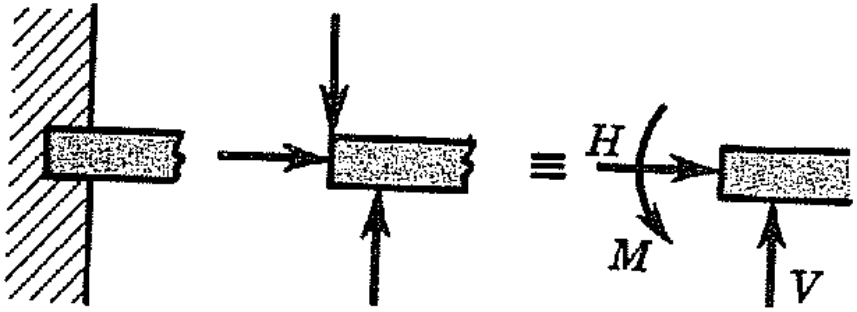


Constraints,action and reaction

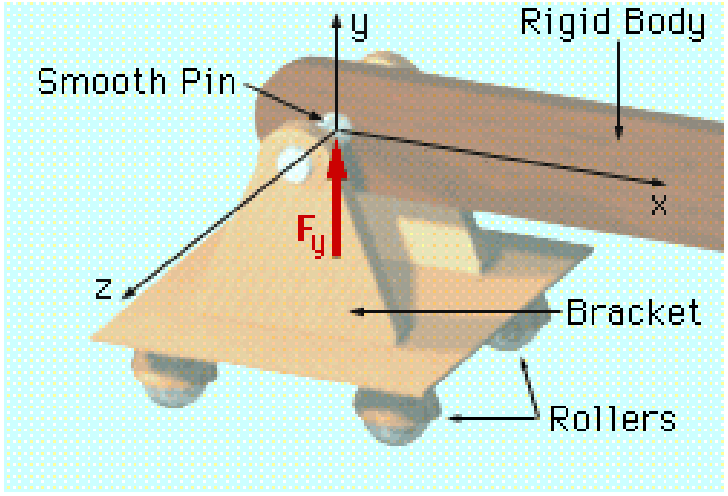
***Constraint**:* Restriction to the free motion of body in any direction.

Based on the nature of constraints the supports are classified as:

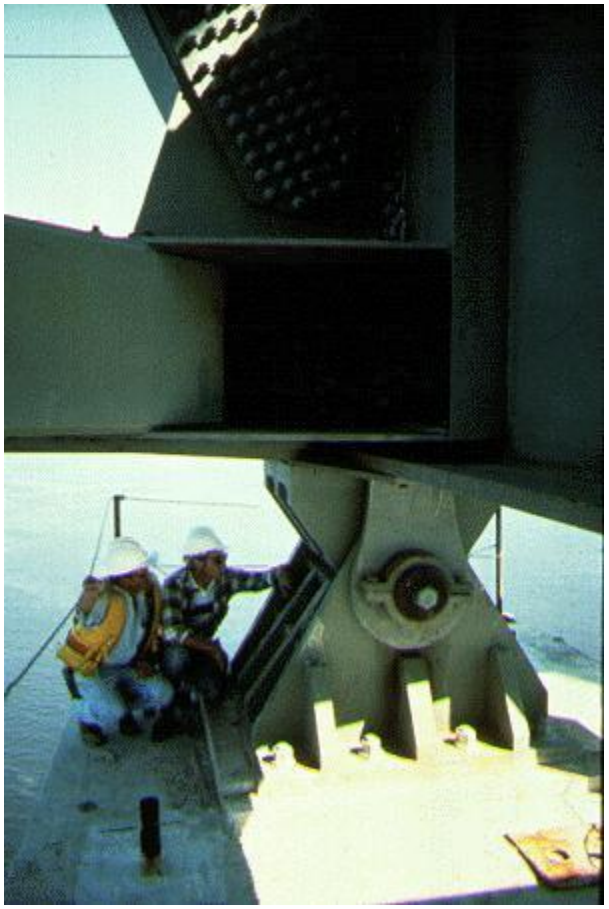
- Roller support
- Hinged support
- Built in support, clamped or fixed



Fixed support



Roller support



Hinged support

Free body diagram

To draw the free body diagram of a body we remove all the supports and replace them by reactions which these supports exerts on the body.

Equations of Equilibrium

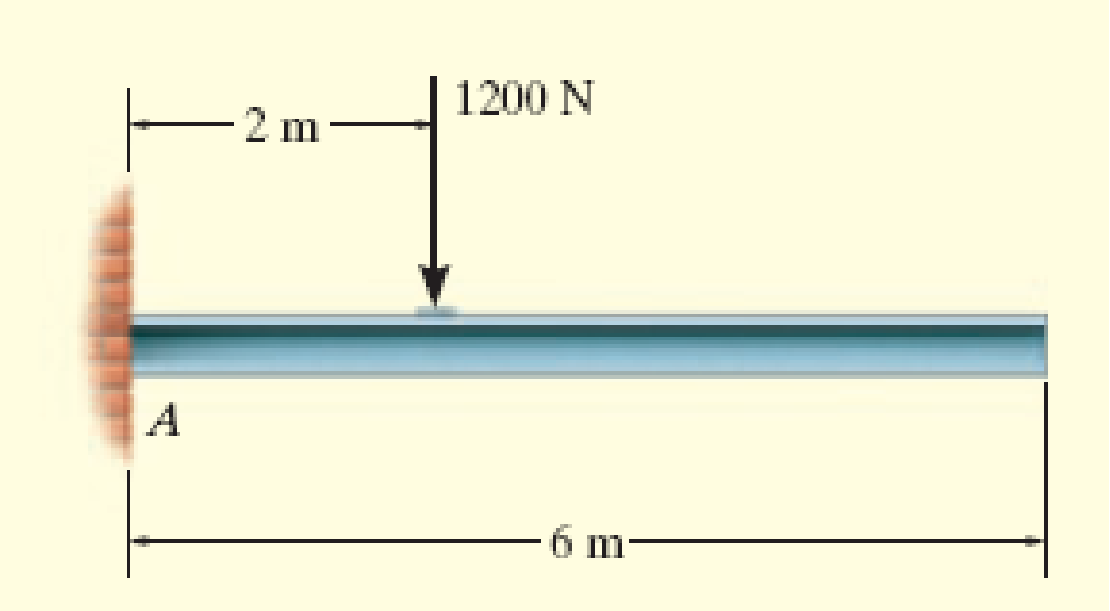
- For equilibrium of a rigid body in 2D,

$$\sum F_x = 0; \sum F_y = 0; \sum M_O = 0$$

- $\sum F_x$ and $\sum F_y$ represent sums of x and y components of all the forces
- $\sum M_O$ represents the sum of the couple moments and moments of the force components

Problem2

Draw the free-body diagram of the uniform beam. The beam has a mass of 100kg.



Equilibrium of a body subjected to two forces

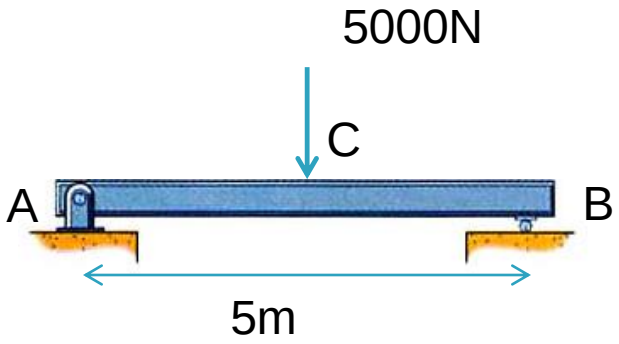
Two forces can be in equilibrium only if they are equal in magnitude, opposite in direction and have the same line of action

Equilibrium of a body subjected to three forces

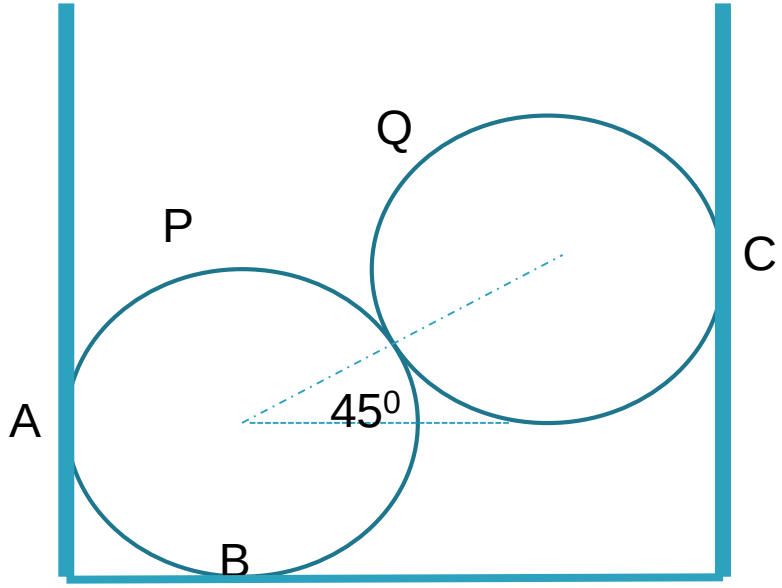
The line of action of all the forces should meet at a point.

Problem3

A force of 5000N is applied at the centre C of the beam AB of length 5m. Find the reactions at the supports.



Problem

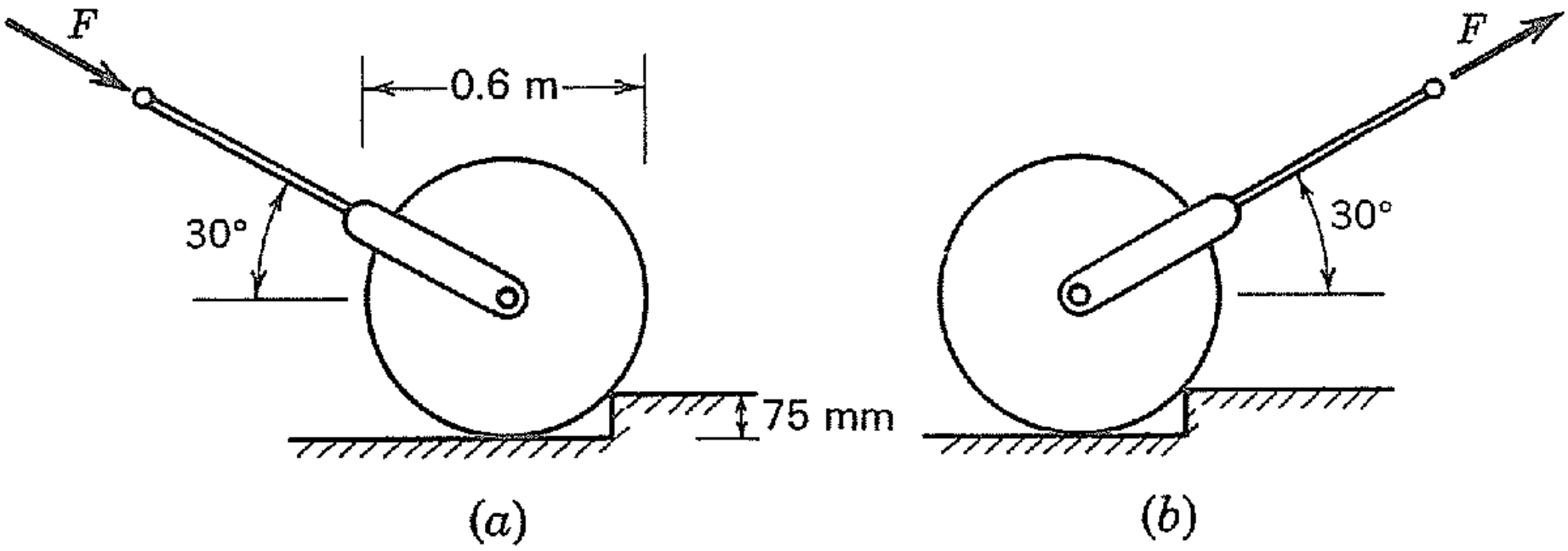


Two similar cylinders P&Q each of weight 900N rests inside a box. Draw FBD of:

a) Both the cylinders taken together
b) The cylinder P
c) The cylinder Q
d) Find the reactions at A,B and C

Neglect frictions between the cylinders and the box

Problem 1.13 TB)



Compare the force F required to just start the 900N lawn roller over a 75mm step when

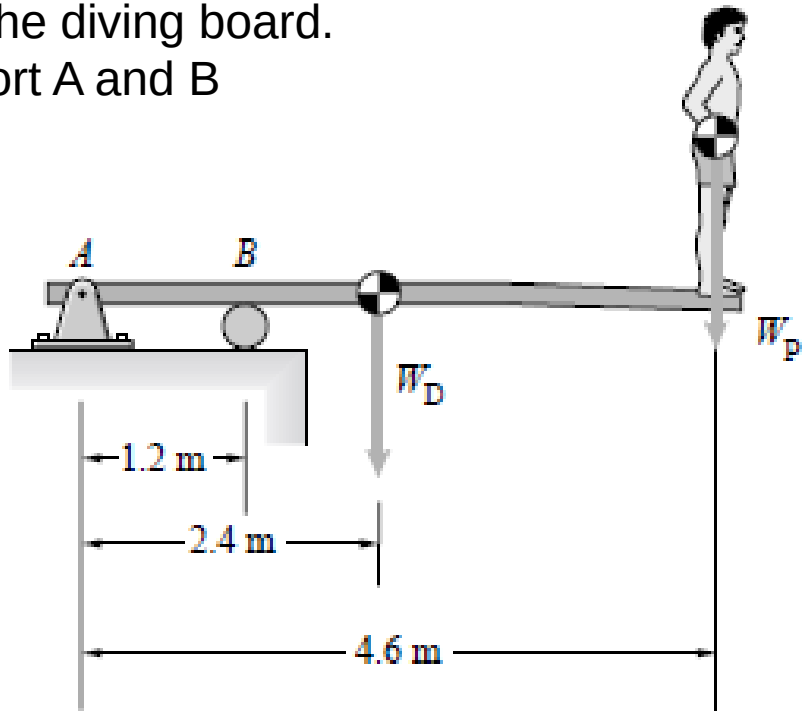
a) The roller is pushed
b) The roller is pulled

Ans: a) 1866.05N b) 607.28N

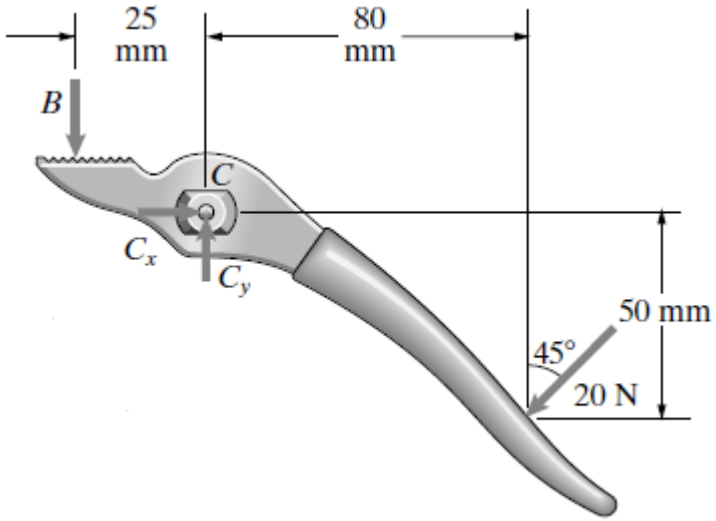
The masses of the person and the diving board are 54kg and 36kg respectively. Assume that they are in equilibrium.

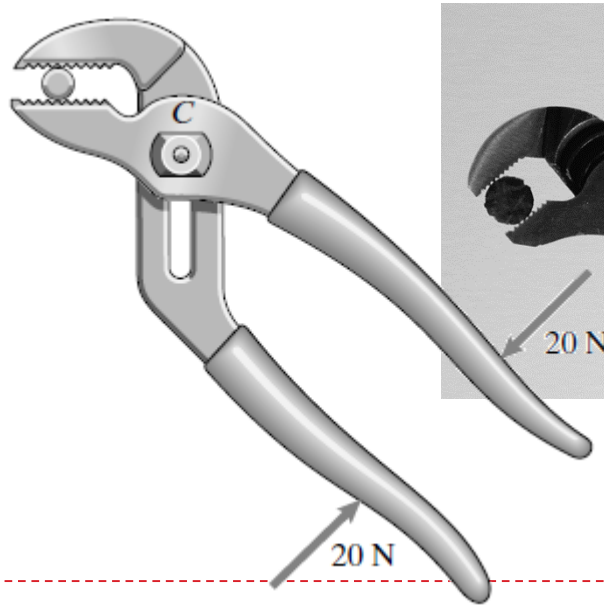
a) Draw the Free body diagram of the diving board.

b) Determine the reactions at support A and B



The person exerts 20-N forces on the pliers. The free-body diagram of one part of the pliers is shown. Notice that the pin at C connecting the two parts of the pliers behaves like a pin support. Determine the reactions at C and the force F_B exerted on the pliers by the bolt.

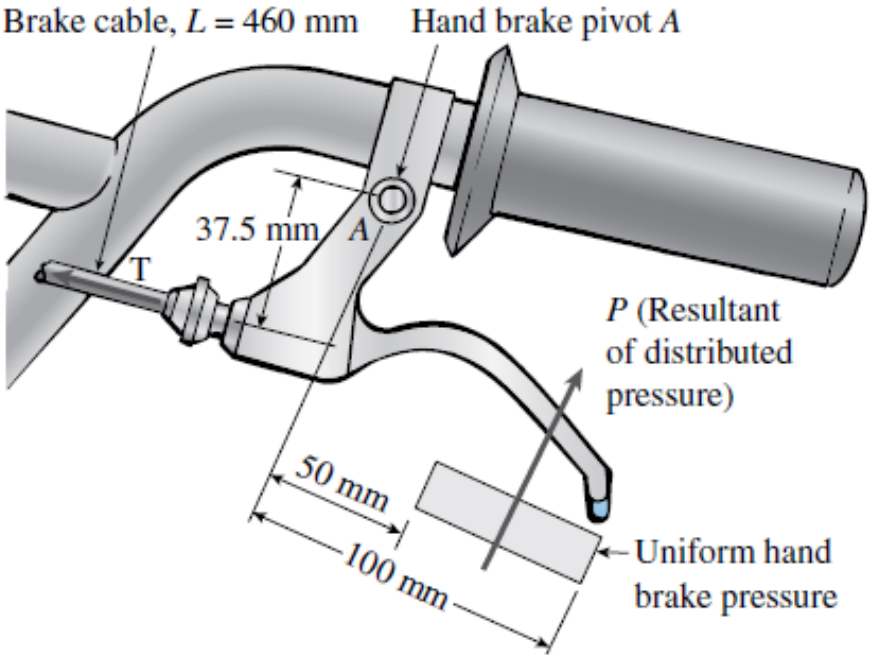




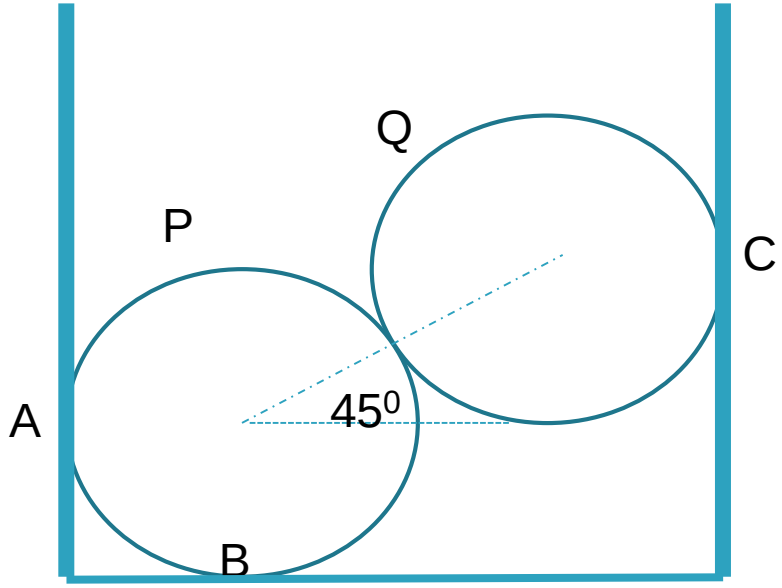


Problem

A force *P of 70 N is applied by* a rider to the front hand brake of a bicycle (*P is* the resultant of an evenly distributed pressure). As the hand brake pivots at *A, a tension T develops in* the 460-mm long brake cable. Find tension T in the brake cable.



Problem



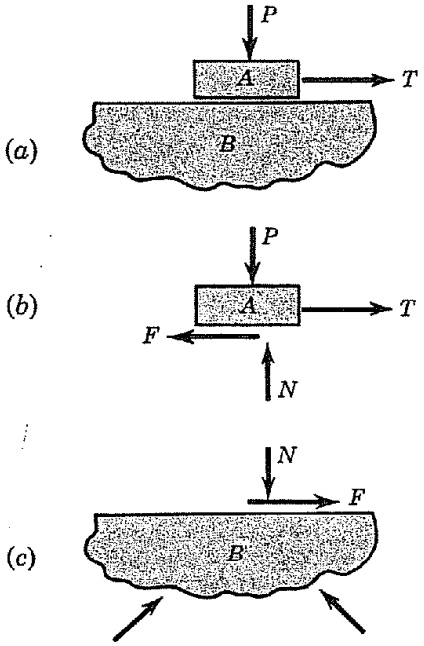
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b) The cylinder P
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Neglect frictions between the cylinders and the box

Friction

- Very important forces of mechanics
- Comes into action whenever a tangential force is applied to a body normally pressed against the surface of the other.



Main properties of friction force acting on A:

- No relative motion between A and B
  - Static friction force, F_S
- Body A slides on body B
  - Kinetic friction force, F_k

Limiting case of friction: $F_S = \mu_S N$

Static Friction Force

- An applied force F attempts to slide a block along a surface
  - If the block does not move,

$$f_s = F$$

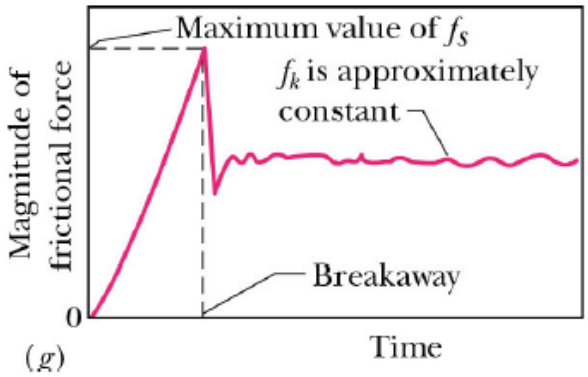
f_s increases when F increases until ….

  - f_s has a maximum value: $f_{s,\,max}$

$$f_{s,\,max} = \mu_s N$$

μ_s : coefficient of static friction

N : normal force



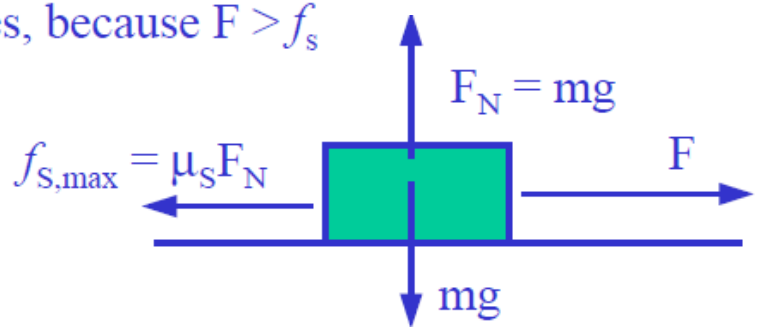
A block lies on a horizontal floor.

a) What is the magnitude of the friction force (f) on it from the floor? $f = 0\text{N}$, but note $f_{S,max} = \mu_S F_N$

b) If a horizontal force of 5 N is now applied to it, but it does not move, what is f now? $f_s = 5\text{N}$

c) If $f_{s,\,max} = 10$ N, will the block move if the horizontal applied force is 8 N? no, because $F < f_s$

d) How about 12 N? yes, because $F > f_s$



Friction and the Normal Force

The maximum frictional force is proportional to the normal contact force.

An increase in the normal force results in an increase in the maximum friction.

This is because the molecules on the two surfaces are pushed together more, thus increasing their interactions.

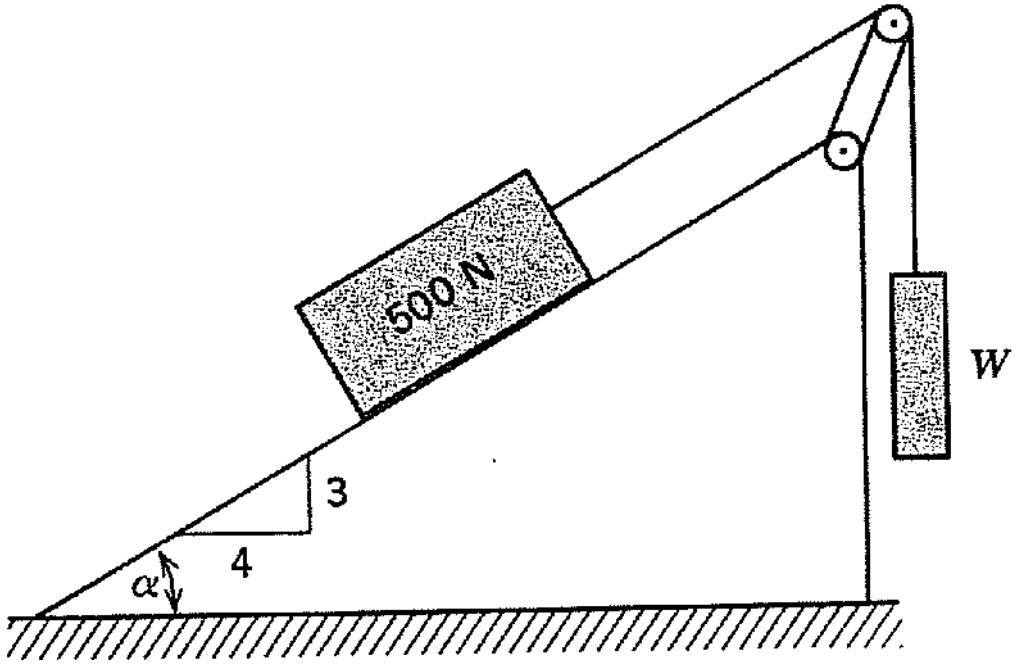
Friction and Surface Area

Friction is not affected by the size of the surface area in contact.

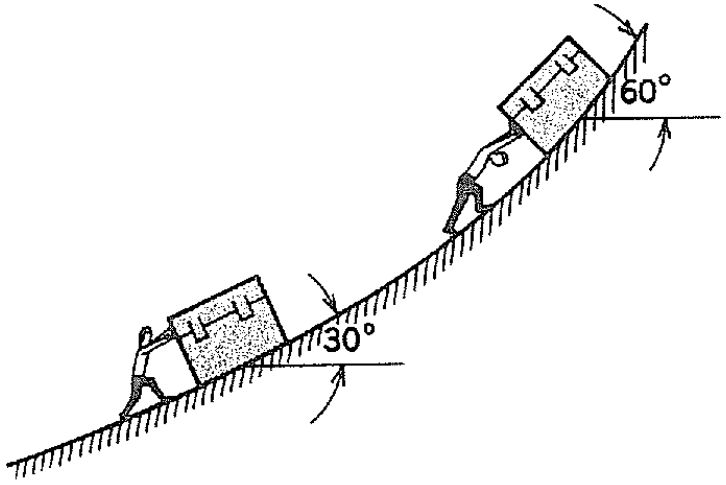
If the normal force remains constant, but the contacting surface area is increased, then the normal force is spread out over more molecules, thus the force on each molecule is reduced.

Find the range of values of W which will hold this block in equilibrium on the inclined plane.

- Assume cable is weightless, pulley is frictionless.
- $f_s = 0.5$

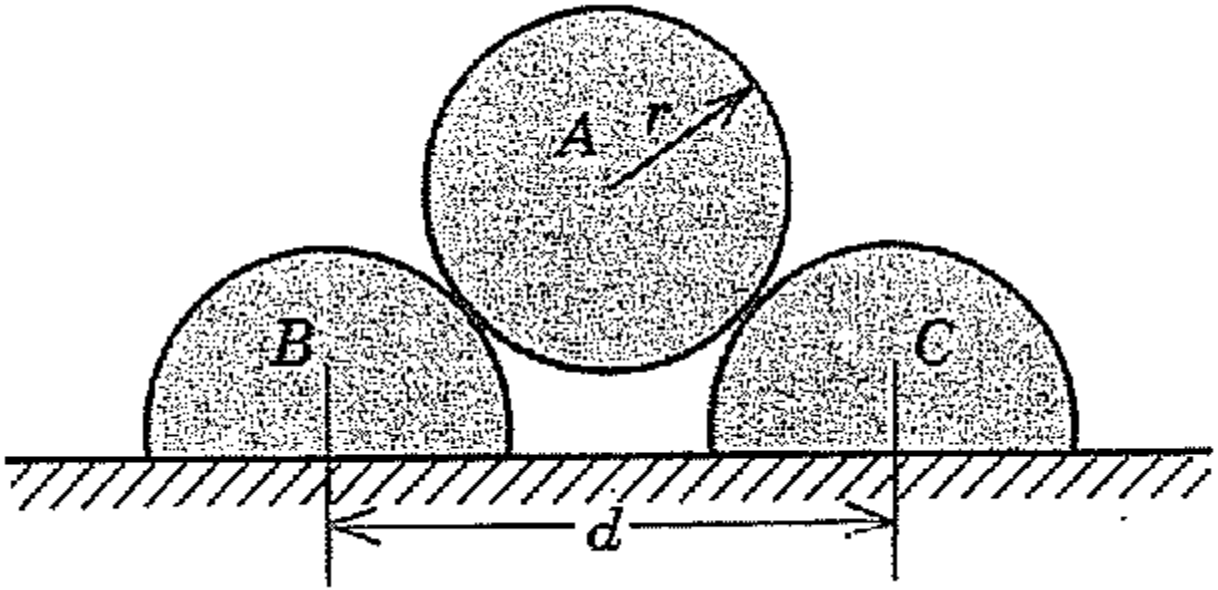


PROBLEM 1.45



A longshore man can barely start pushing a trunk up a 30^0 concrete ramp. He can barely hold it from sliding back when the slope is 60^0 . What is the coefficient of static friction between the trunk and concrete.

PROBLEM 1.37

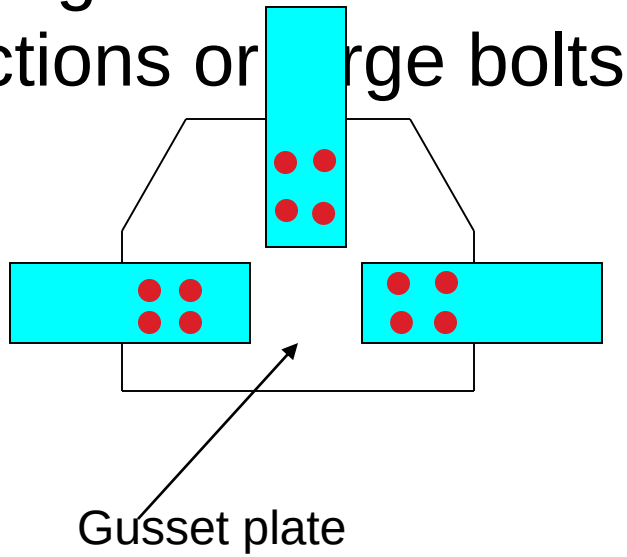


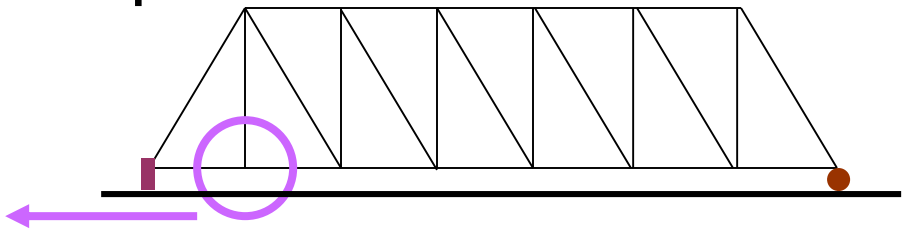
A circular cylinder A rests on top of two-circular cylinders B and C, all having the same radius r. The weight of A is W and that of B and C is W/2 each. Assume that coefficient of friction between the flat surfaces of the half cylinders and the horizontal table top is f. Determine the maximum distance d between the centres of the half cylinders to maintain equilibrium.

TRUSSES

Trusses: Engineering structures that are composed only of *two-force members. e.g., bridges, roof supports*

Actual trusses: Airy structures composed of slender members (I-beams, channels, angles, bars etc) joined together at their ends by welding, riveted connections or large bolts and pins





A typical truss structure

- All loads are applied / transmitted at joints
- All members are joined by pin connections
- **Consist entirely of two-force members**

- **Can contain zero-force members**

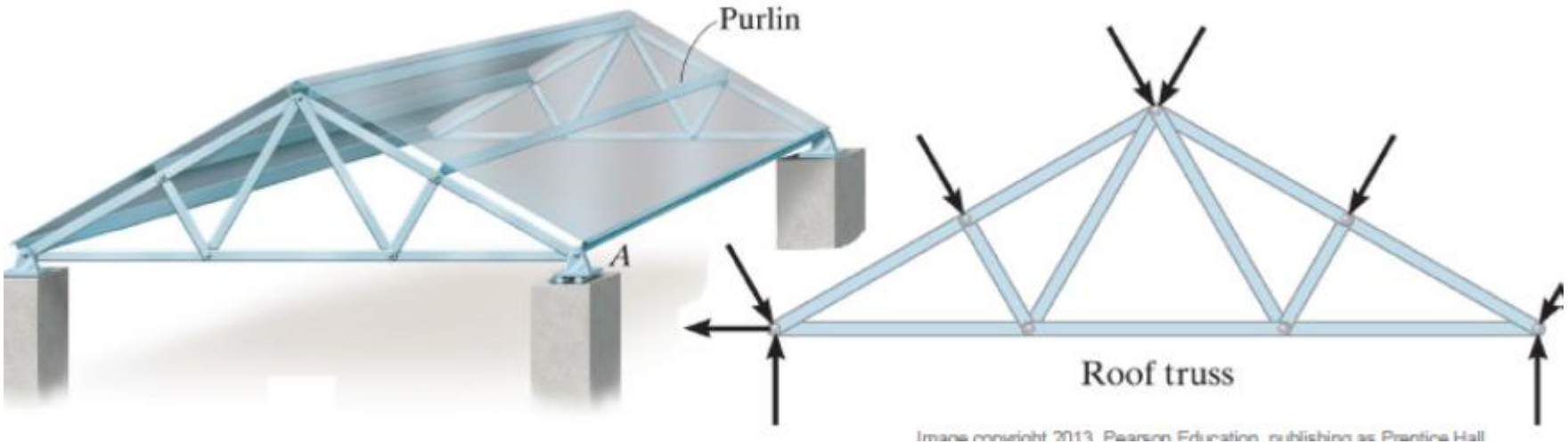
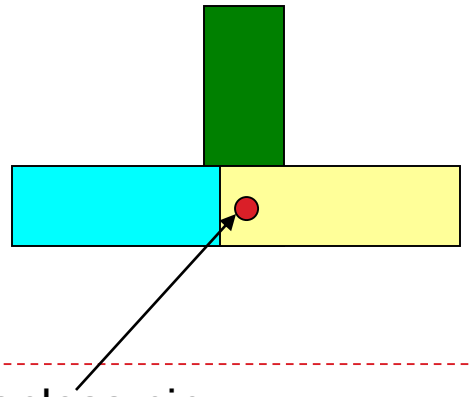


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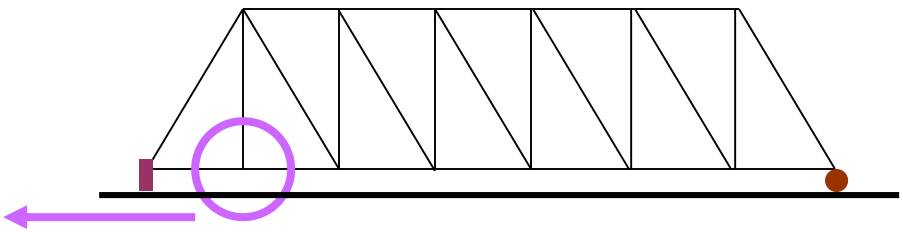
Ideal trusses:

Assumptions

- Ideal truss members are connected only at their ends.
- Ideal truss members are connected by frictionless pins
- The truss structure is loaded only at the pins(joints)
- Weights of the members are neglected

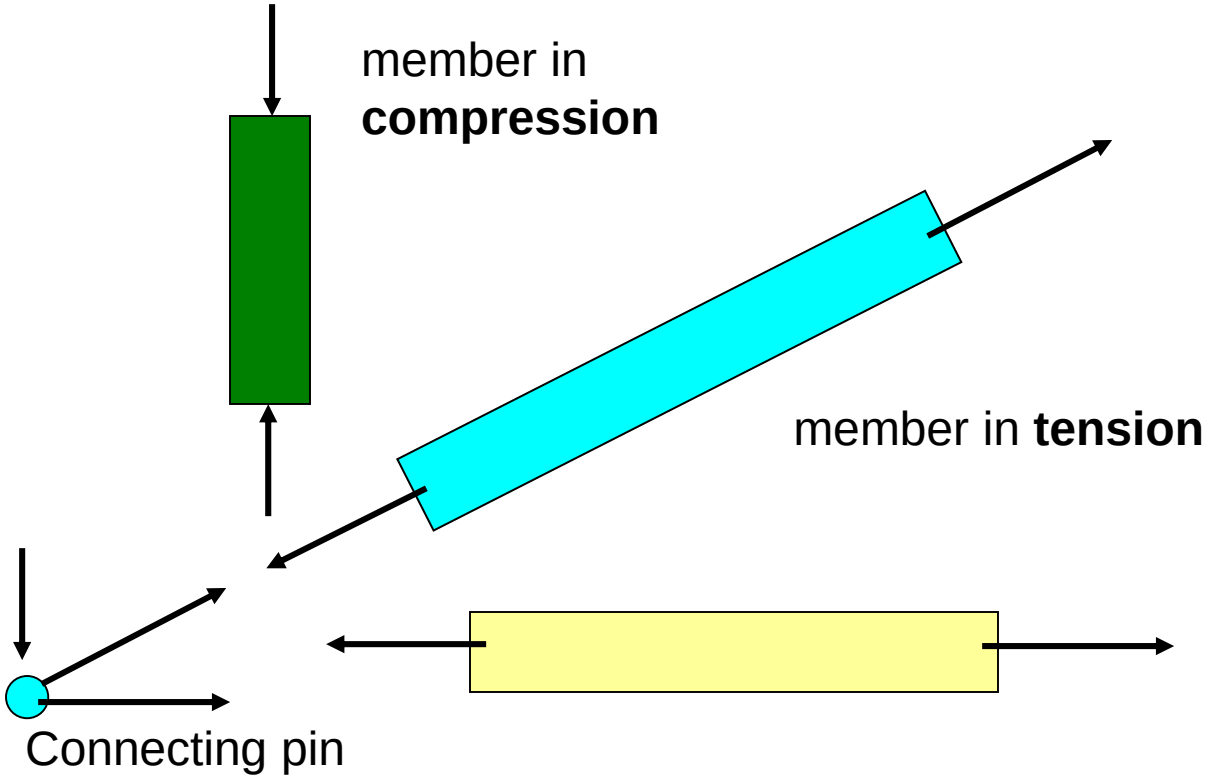


Frictionless pin



A typical truss structure

These assumptions allow us to idealize each truss member as a two-force member (members loaded **only** at their extremities by equal opposite and collinear forces)



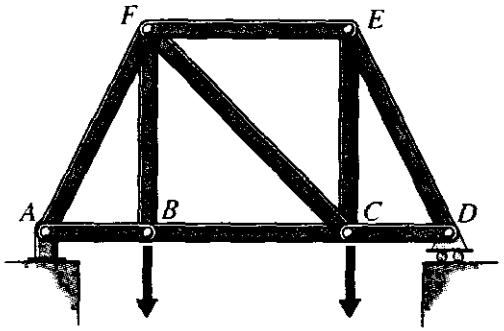
Stability Criteria

m=2j-3 (rigid and perfect truss)

j- number of joints.

m- number of members.

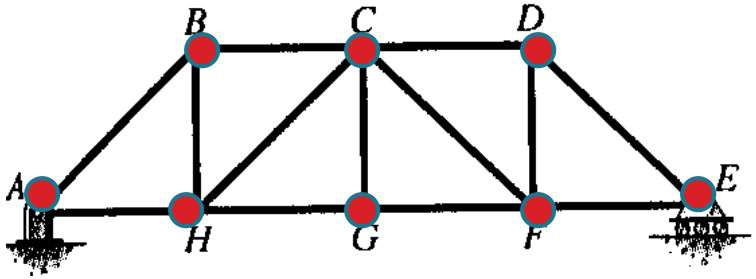
3- number of support reaction



m<2j-3 → Truss unstable (under rigid and collapsible)

m>2j-3 → Over rigid and Statically indeterminate

Example



m (Number of members) =13

j (Number of joints) = 8

Number of supports reactions=3

m=2j-3

Analysis Methods (to determine axial force in members)

innovate achieve lead

Methods of Joints

Method of Sections

Method of Joints

Truss in Equilibrium => Each Joint in Equilibrium

Procedure

STEP1: Calculate support reactions

Consider the entire truss (structure) as a rigid body and draw FBD**.** Then using eq. of equilibrium calculate the support reactions.

STEP 2: Select a joint

At which there are max two unknown forces and one or more known forces acting on it. Assume the forces in all the members to be tensile. Draw FBD of the joint and apply eq of equilibrium to it.

STEP 3:SELECT OTHER JOINTS

Repeat the step 2 until all the forces in the members are calculated.

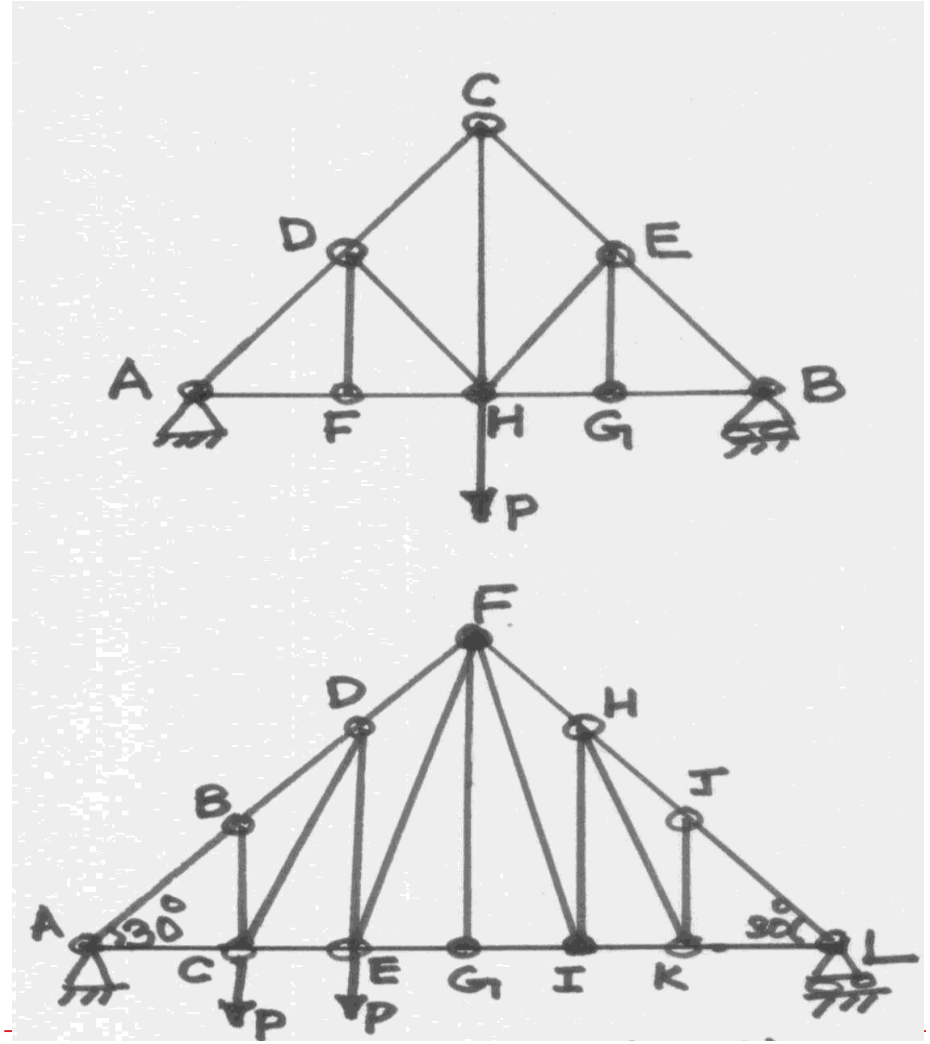
STEP 4: Write the answers in table representing members, forces, nature (compressive or tensile)

Zero Force Members

These members are used to increase the stability of the truss during construction and to provide support if the applied loading is changed. There are two conditions:

1. When 2 members meeting at a joint are not collinear and there is no external load or support reaction acting at the joint, the members must be zero force members.
2. When there are 3 members meeting at a joint, of which 2 are collinear and the third be at an angle and if there is no external load or support reactions at the joint the force in the third member is zero.

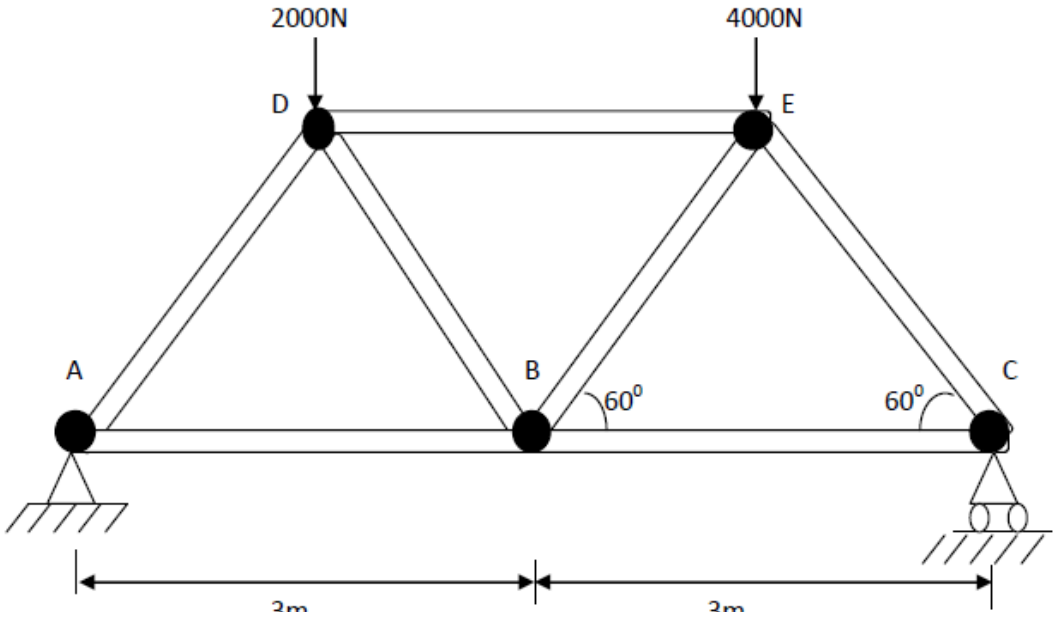
Identify the zero force members



F_{FD}, F_{GE}

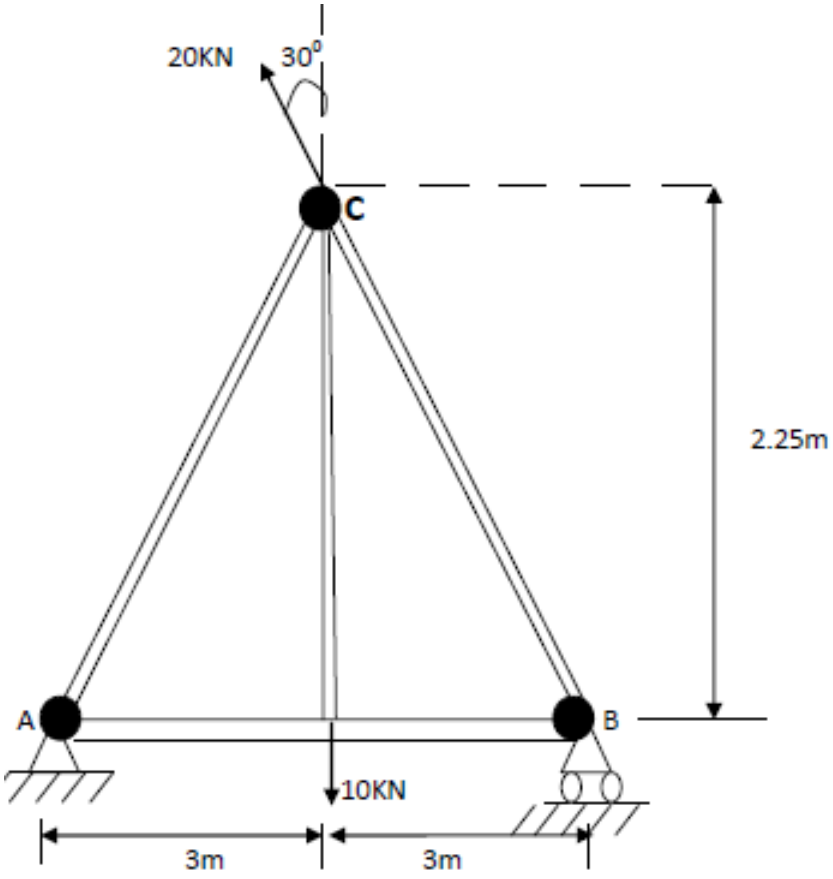
F_{BC}, F_{FG}, F_{FI}, F_{HI}, F_{HK}, F_{JK}

Calculate the support reactions and axial forces in all the members



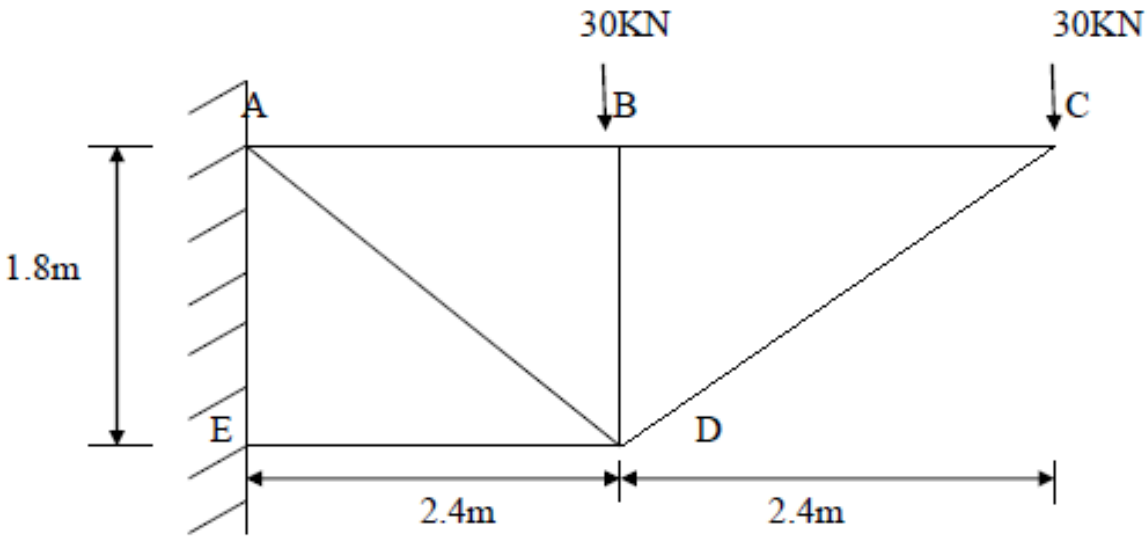
MEMBER	FORCES(N)	NATURE
AB	1443.375	T
BC	2020.75	T
AD	2886.75	C
BD	577.36	T
BE	500	C
CE	4041.45	C
DE	1732.055	C

Calculate the support reactions and axial forces in all the members



S.NO.	Member	Forces(KN)	Nature
1.	AB	16.52	C
2.	AD	23.21	T
3.	CD	10	T
4.	BD	23.21	T
5.	BC	29	C

Find the forces in members of the truss



Since it is a cantilever truss calculations of reactions are not necessary

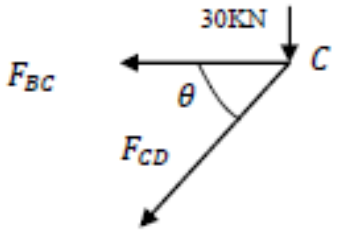
JOINT C

$\tan\theta = 1.8/2.4 =$

$\theta = \boxed{36.86^0}$

$\Sigma F_x = 0$

$F_{BC} + F_{CD}\cos 36.86 = 0$



$\Sigma F_Y = 0$

$30 + F_{CD}\sin 36.86 = 0$

$\boxed{F_{CD} = 50.01\text{KN(C)}}$

$F_{BC} = -(-50.01\cos 36.86) = \boxed{40.01\text{(T)}}$

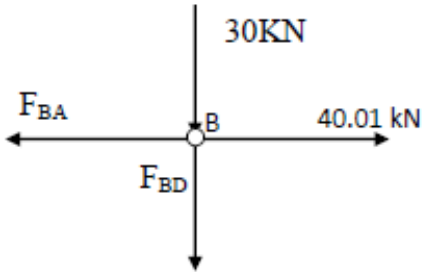
JOINT B

$\Sigma F_x = 0$

$F_{BA} = 40.01KN(T)$

$\Sigma F_Y = 0$

$F_{BD} = 30KN(C)$



POINT D

$\Sigma F_x = 0$

$F_{DA} \cos 36.86 + F_{DE} + F_{CD} \cos 36.86 = 0$

$\Sigma F_Y = 0$

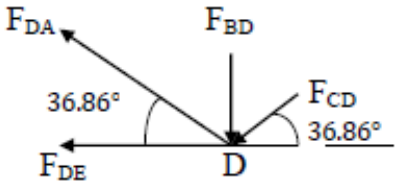
$F_{DA} \sin 36.86 = 30 + F_{CD} \sin 36.86$

$F_{DA} = 100.02(T)$

$0.800 + F_{DE} + 40.01 = 0$

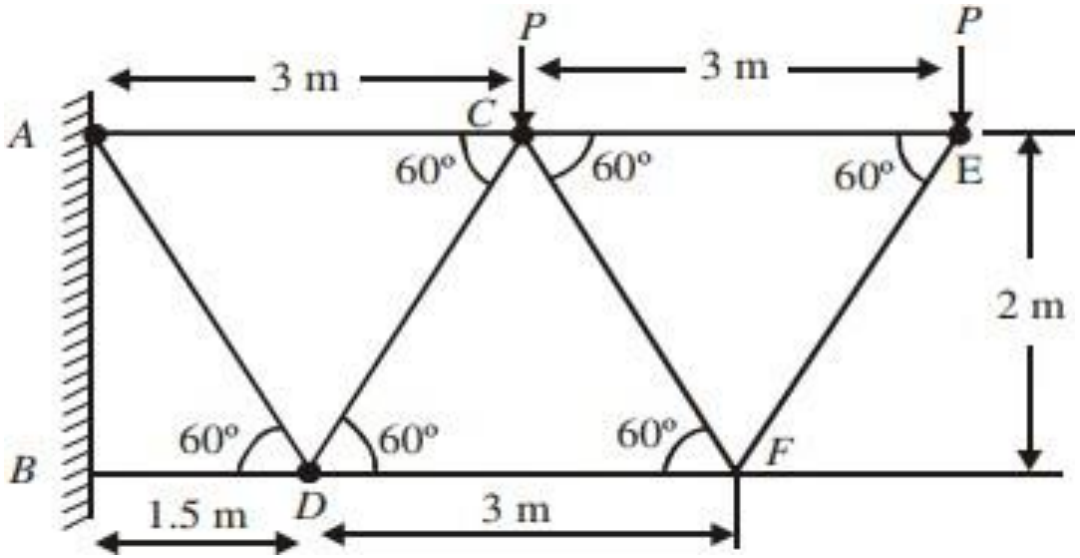
$F_{DE} = -120.03KN$

$F_{DE} = 120.03KN(C)$



QUESTION2

A cantilever truss is loaded and supported as shown in fig. find the values of load P which would produces an axial forces of magnitude 3KN in the member AC



S.NO.	MEMBERS	FORCES(KN)	NATURE
1.	CB	40.01	T
2.	CD	50.01	C
3.	BD	30	C
4.	BA	40.01	T
5.	DA	100.02	T
6.	DE	120.03	C

Method of sections

The method of joints is most effective when the forces in all the members of a truss are to be determined. If however, the force in only one or a few members are needed, then the method of sections is more efficient.

Step1:calculate support reactions if necessary

Step 2: pass an imaginary section, which cuts max 3 members including the members in which the forces are required to be calculated. The section should separate the truss entirely in to two portions.

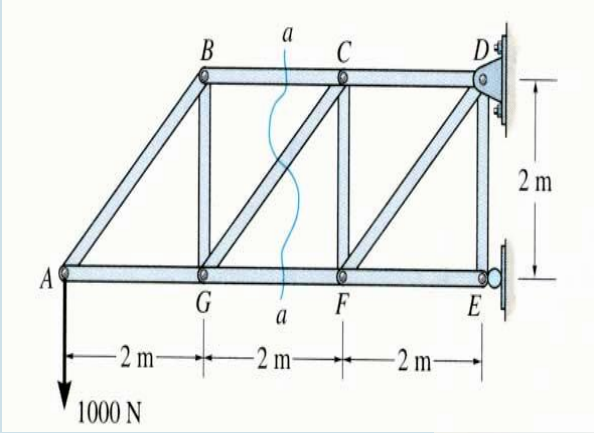
* the section should pass through the members not through the joints.

Step3: choose the portion with minimum no. of forces. (if same any portion could be chosen)

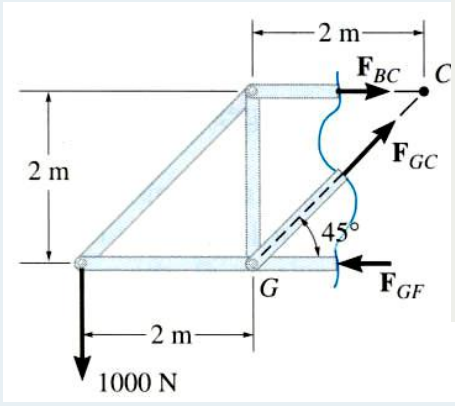
Step 4: for further calculations the forces of only chosen portions has to be considered. But the moment equilibrium equations could be applied at any point of the truss.

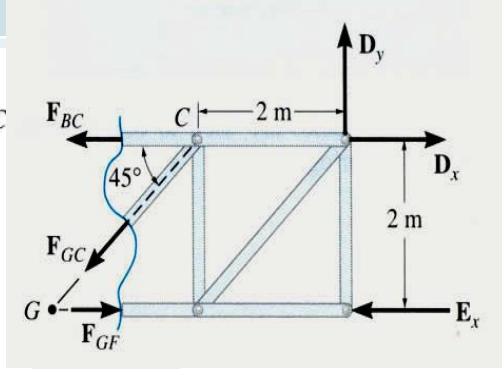
Step I :calculate support reactions if necessary

Step 2: pass an imaginary section, which cuts max 3 members including the members in which the forces are required to be calculated. The section should separate the truss entirely in to two portions

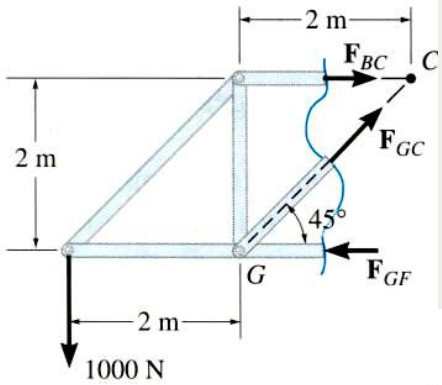


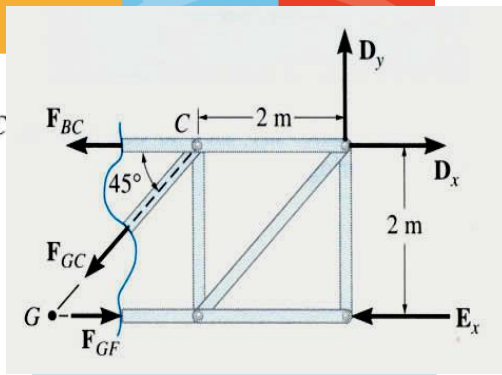
* Draw FBD of two portions separated. Show the unknown forces in the members(we will assume all the forces in the members to be tensile to start with)

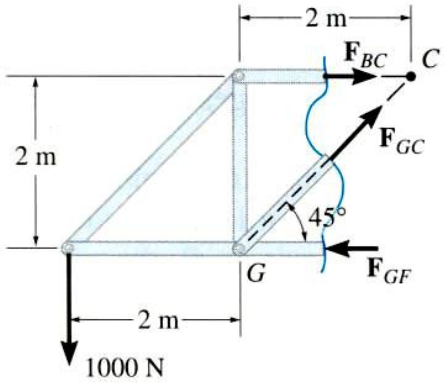




Step3: choose the portion with minimum no. of forces. (if same any portion could be chosen)





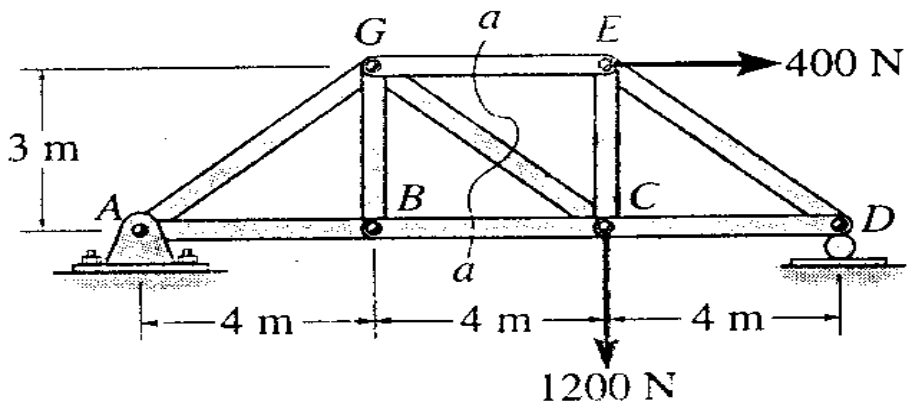


Step 4: for further calculations the forces of only chosen portions has to be considered. But the moment equilibrium equations could be applied at any point of the truss.

Apply the three equations of equilibrium to the section.

- **If possible, attempt to develop an equation in just one unknown.**
- **Look for points where the lines of action of several forces are concurrent**

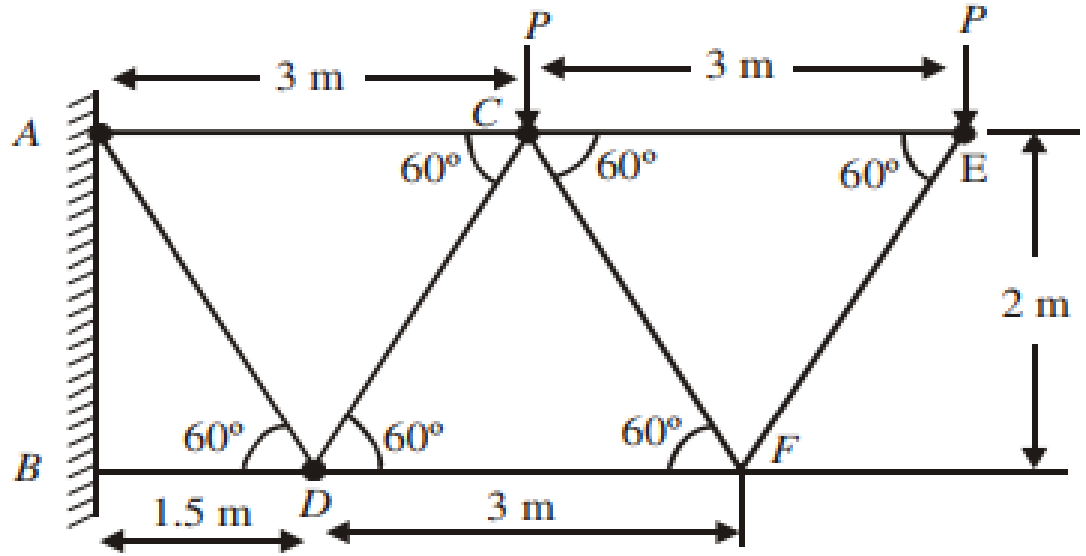
Determine the force in members GE, GC and BC of the truss shown in the Figure. Indicate whether the members are in tension or compression.

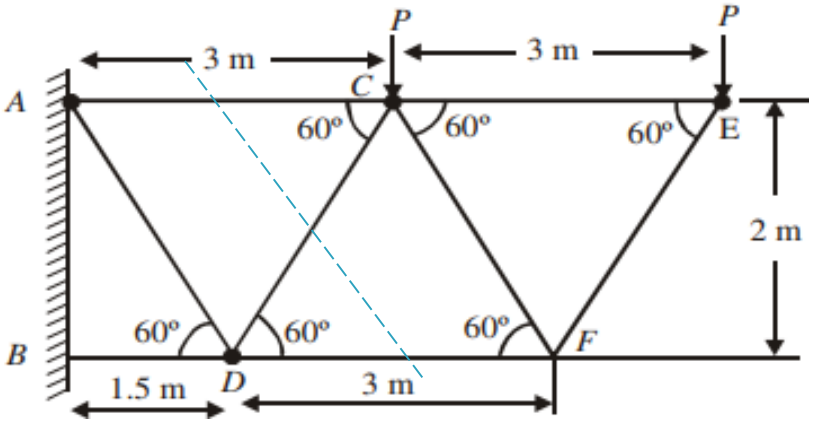


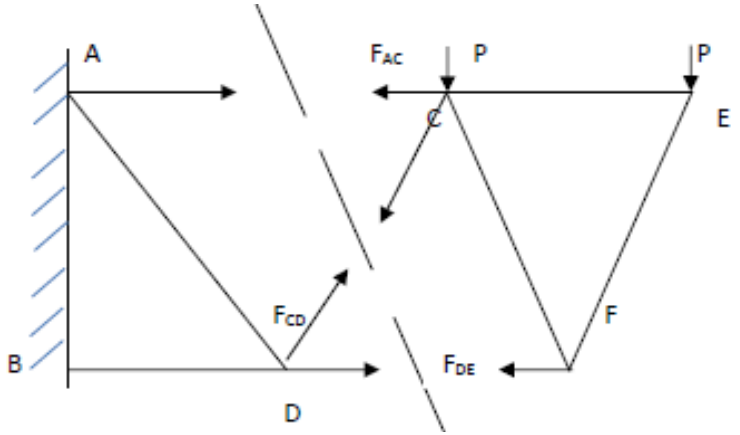
(a)

ANS:
RAx=400N(←)
RAy=300N
RD=900N
FGE=800N(C)
FBC=800N(T)
FGC=500.11N(T)

A cantilever truss is loaded and supported as shown in fig. find the values of load P which would produces an axial forces of magnitude 3KN in the member AC







$+\circlearrowleft \Sigma M_D = 0$

$F_{AC} \times 2 - P(1.5) - P(4.5) = 0$

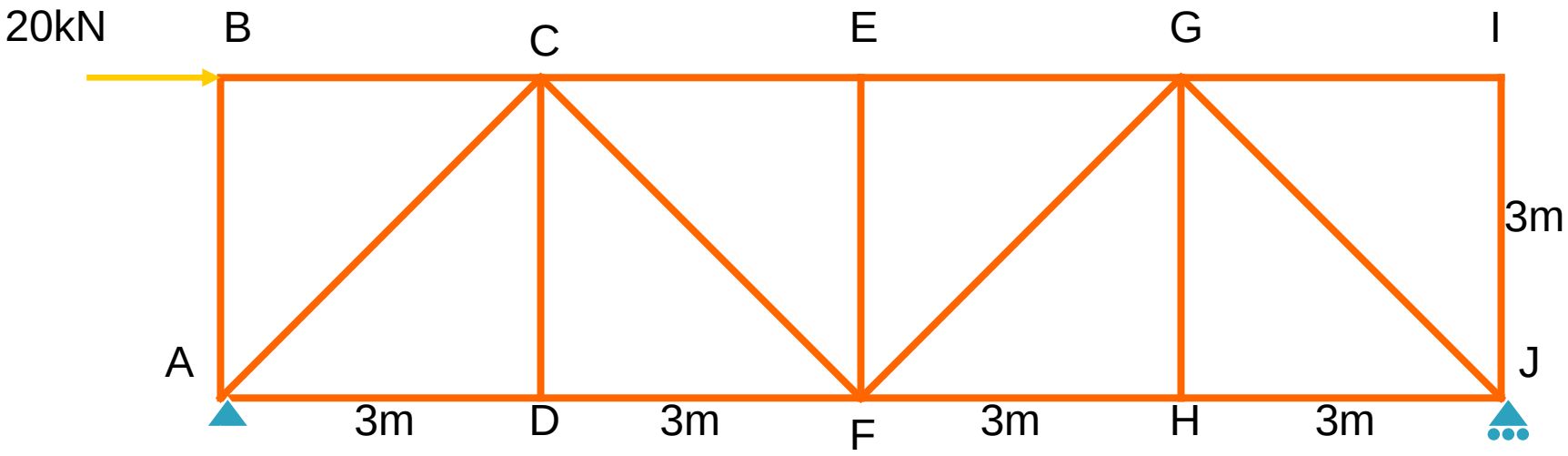
$\boxed{F_{AC} = 3P}$

As force in member AC is 3KN

$3P = 3$

$P = 1KN$

Find the forces in members CE, CF and FH of the truss shown below.

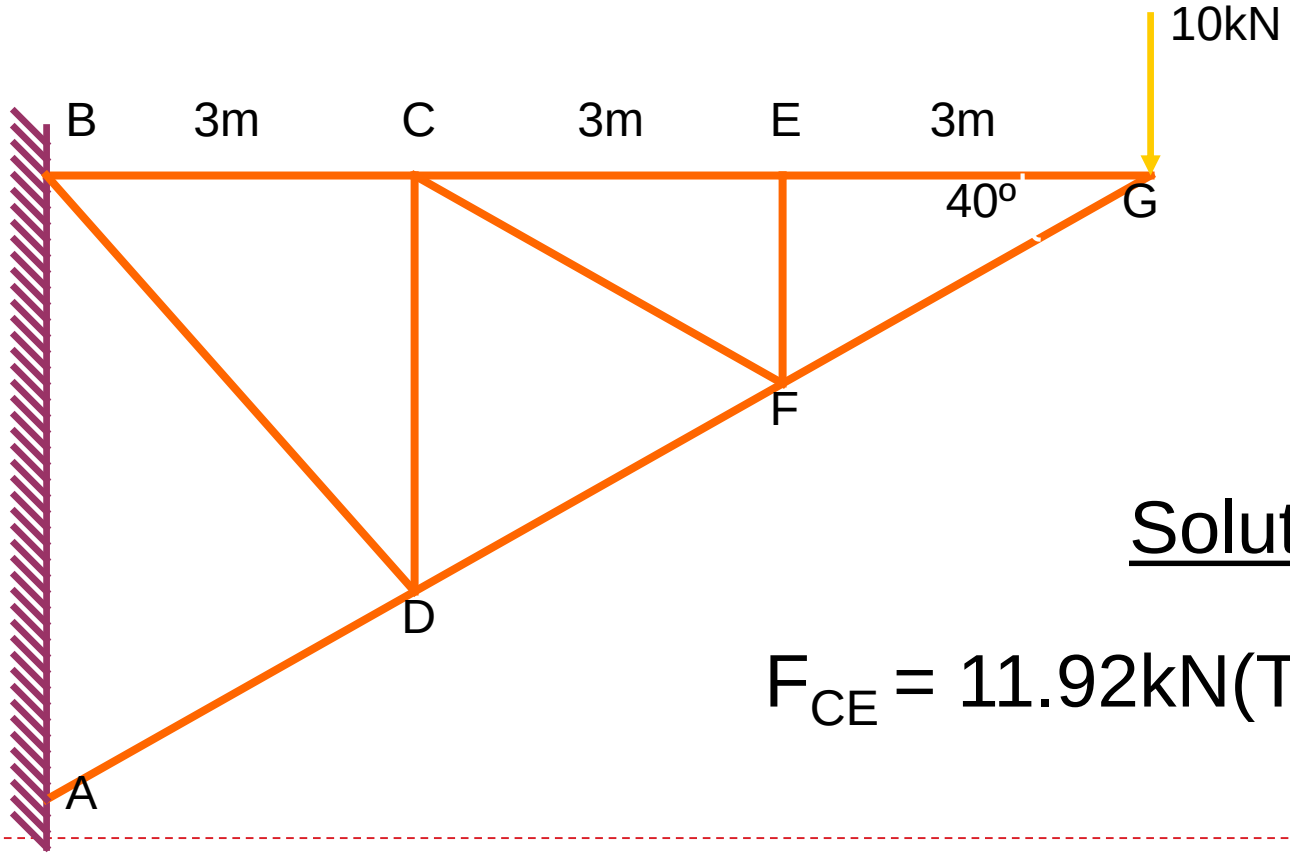


Solution : $F_{CE} = 10kN(C)$

$F_{CF} = 7.07kN(C)$

$F_{FH} = 5kN(T)$

Find the forces in members CE and CD of the truss shown below.



Solution :

$F_{CE} = 11.92kN(T)$ $F_{CD} = 0$


THANK YOU FOR YOUR ATTENTION